

ADITYA GIRISH

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EDUCATION

Bachelor of Technology (B.Tech.), Computer Science, 2022-2026 PES University, Bengaluru CGPA: **8.71**
Honors & Scholarships: MRD Scholarship - Top 5% ; CNR Scholarship x 2 - Top 10%

PUBLICATIONS

MSTRAE: Multi-Scale Temporal Recurrent Autoencoder for Flight Manoeuvre Recognition

Lead Author - Accepted - Future Technologies Conference (FTC 2026)

Springer Lecture Notes in Networks and Systems

Models Cessna 172 flight telemetry with a multi-scale LSTM autoencoder to learn hierarchical temporal representations for zero-shot manoeuvre recognition. Clusters latent manoeuvre representations using HMM-based unsupervised inference and evaluates recognition without hand-labelled ground truth via an LLM-based assessment framework.

When Does Structure Help? Graph vs. Tabular Models for Agent-Workflow Cost Prediction

AAAI Conference on Artificial Intelligence (AAAI 2027) Under Review **Main Technical Track** *Solo Author*

Benchmarks graph neural networks against tabular learners over 357K coding-agent rounds for execution-cost prediction under structured workflow dependencies. Finds that message-passing graph models degrade when cost signals do not propagate through the execution graph, establishing when structural inductive bias is uninformative.

Execution State Attribution: From Provenance Records to Causally Sufficient State in Agent Runs

NeurIPS 2026 Workshop on Attribution (ATTRIB)

Submitted Methods Paper

Solo Author

Formalizes execution state attribution as recovery of minimal γ -sufficient subsets of runtime state whose resampled complements preserve the observed agent behavior. Shows that per-step necessity scores cannot recover jointly sufficient causes under conjunctive dependencies and attenuate redundant independent causes, motivating set-level causal attribution.

EXPERIENCE

TruEstate

Bengaluru, India

AI/ML Engineer

Jul 2026 - Present

- Specified the full voice-AI stack for an outbound cold-calling pipeline (**telephony, STT, LLM, TTS**), setting per-hop latency budgets and barge-in behaviour so callers interrupt mid-utterance without losing turn state.
- Replaced static call scripts with **per-lead context curation** that joins CRM history and live listing data at call time, so each agent opens on the properties and prior interactions specific to that lead.
- Automated **Meta ad creative production** end to end, building a pipeline that renders image and video variants directly from structured listing fields instead of a per-listing design request.

Software Engineering Intern - AI

Jan 2026 - Jul 2026

- Collapsed a **three-agent design into one classification-routed agent** spanning sales, service-delivery, and post-sales, using tool calling for user-data retrieval and hierarchical vector search; slated for production.
- Cut broker site-build time from **2 hours to 5 minutes** and lifted lead quality **60%** with an agent-based CMS on GCP generating microsites in one click, with Cloudflare deploys, OTP verification, and CRM lead routing.
- Pitched AI capabilities to **500+ brokers across two industry events**, translating model behaviour into broker-level business value and gathering the requirements that shaped subsequent product iterations.
- Shipped an image-generation pipeline on **Gemini 2.5 Flash Image** that virtually staged interiors across **200+ property listings**, removing manual design turnaround from the listing-publish path.
- Originated and demoed a **2D-to-3D reconstruction pipeline** turning architectural floor plans into interactive **Three.js** models, using a **Gemini VLM** for layout-element detection and spatial parsing.

AI Backend Engineer Intern

Apr 2025 - May 2025

Viable Ideas Private Limited

Bengaluru, India

- Deployed an **LLM-based** inventory prediction model to production at **85% classification accuracy** on transaction data from 4 bank sources; automated multi-source ingestion cut manual workflows **70%** and cost **40%**.

PROJECTS

QRAG : Quantized Bi-Level Retrieval Augmented Generation

- Achieved **95.1% recall@10** on FIQA at a **3-4x memory reduction** by quantizing the coarse and fine retrieval tiers independently in a bi-level index built for memory-constrained retrieval.
- Beat **FAISS IVF-PQ** by **21.5pp in recall** at comparable memory budgets across a 57,000-document benchmark, isolating where bi-level quantization wins over standard ANN index compression.